

PROBLEM STATEMENT

Customer Support Ticket Classification Using NLP and Machine Learning

Industry Context

In today's digital world, customer support has become an important part of almost every organization. E-commerce companies, banks, telecom providers, software companies, hospitals, educational institutions, and government service portals all depend on support teams to answer customer questions, resolve complaints, and maintain customer satisfaction. The growth of online services, mobile applications, and digital communication channels has greatly increased the number of support requests that organizations receive every day. These requests arrive through emails, web forms, chat systems, social media, and helpdesk platforms.

Customer support tickets can cover many different topics. Customers may report billing errors, ask for refunds, request password resets, complain about delivery delays, report technical faults, ask for product information, or request changes to their accounts. Some tickets are simple and routine, while others are urgent and require immediate attention. If tickets are not understood and routed correctly at an early stage, they can cause serious problems for the organization.

Poor handling of support tickets can result in several problems, such as:

- Delayed responses and longer resolution times
- Tickets assigned to the wrong department or support agent
- Urgent or critical issues remaining unnoticed in a large queue
- Repeated transfers of the same ticket between teams
- Customer dissatisfaction and loss of customer trust
- Increased workload and stress for support agents
- Higher operational costs due to manual sorting of tickets
- Inconsistent categorization of similar issues
- Difficulty in identifying common problems and recurring complaints

Many organizations use helpdesk software, keyword-based rules, and manual triage by support staff to organize incoming tickets. These methods are useful for simple and predictable requests, but they have difficulty handling ticket text that is informal, ambiguous, misspelled, or written in

different styles. Rule-based systems also require constant manual updates and may fail when customers describe the same problem in different words.

Another challenge is the increasing volume of tickets. Manually reading and sorting every ticket is not practical when an organization serves a large number of customers. Support managers need a system that can automatically understand the content of each ticket, identify its category, and judge how urgent it is.

Natural Language Processing (NLP) and Machine Learning can be used to address this problem by learning patterns from previously resolved tickets and using them to understand new ones. An intelligent classification system can read the ticket text, recognize the type of issue, estimate its priority, and route it to the right team without manual effort.

Therefore, there is a need for an AI-based support solution that provides automatic ticket understanding, faster routing, consistent categorization, and clear visualization of support activity. Such a system can help organizations respond to customers more quickly and use their support resources more effectively.

Problem Statement

Design and develop a Customer Support Ticket Classification System Using NLP and Machine Learning that automatically analyzes incoming support tickets and classifies them into suitable categories and priority levels.

The system should collect and analyze relevant ticket information such as ticket subject, ticket description, customer details, submission channel, product or service name, date and time of submission, previous ticket history, and other important attributes. The collected data should be processed before analysis so that unnecessary, incomplete, duplicated, or inconsistent information does not affect the classification process. This includes text cleaning, tokenization, stop-word removal, handling of spelling errors, and conversion of text into numerical features.

The system should use suitable NLP and Machine Learning techniques to understand the meaning of each ticket and identify the type of issue being reported. It should be capable of classifying tickets into categories such as billing, technical support, account access, delivery,

refunds, product inquiry, and general feedback. When a ticket is classified, the system should also assign an appropriate priority or urgency level based on its content and potential impact.

The system should automatically route classified tickets to the appropriate team or support agent and notify them in a timely manner. A monitoring dashboard should display useful information such as the number of tickets received, category distribution, priority level, average classification confidence, routing status, and current queue status. This will help support managers understand the situation without having to manually read large numbers of tickets.

The system should also maintain historical records of processed tickets. These records can be used to understand previous support activity, identify recurring issues, analyze customer complaint trends, and study changes in ticket volume over time. The system should provide simple reports and visualizations so that managers can easily review support performance information.

The proposed solution should be designed to support continuous operation and should be capable of handling increasing numbers of tickets. It should also consider important requirements such as secure data handling, reliability, scalability, efficient processing, and timely routing of tickets. The overall aim is to create an intelligent support system that reduces manual effort, improves response time, and increases customer satisfaction.

In addition, the system should provide a clear method for prioritizing tickets based on their urgency and potential business impact. Low-priority general inquiries should be distinguished from critical issues such as service outages, payment failures, or security-related complaints, so that agents can focus on tickets that require immediate attention. The system should also maintain an audit trail of processed tickets, including the time of receipt, predicted category, assigned priority, confidence score, assigned team, and any corrections made by the support agent. This information can be used to review previous decisions, find misclassified tickets, and improve the model through feedback. The system should be designed with proper access control and secure handling of customer information to prevent unauthorized access to sensitive personal data.

Objectives

The system should:

- To continuously collect incoming customer support tickets from multiple channels.
- To monitor ticket queues and support activity in real time.
- To clean and prepare ticket text and related data before performing classification.
- To identify the main issue and intent expressed in each ticket.
- To classify tickets into predefined categories accurately and consistently.
- To use NLP/ML techniques to improve the understanding of customer messages.
- To assign priority or urgency levels based on ticket content and impact.
- To route tickets automatically to the correct team or agent.
- To generate real-time alerts when high-priority or critical tickets are received.
- To provide a scalable system that can handle increasing ticket volume and support data.
- To improve the organization's ability to respond to customer issues quickly and effectively.

Expected Outcomes

1. Develop an end-to-end Customer Support Ticket Classification System for collecting, preprocessing, classifying, and routing support tickets using NLP and Machine Learning.
2. Provide a GUI/dashboard for real-time monitoring, visualization, and analysis of ticket volume, categories, priorities, and routing status.
3. Maintain information such as ticket ID, customer details, ticket subject and description, predicted category, priority level, confidence score, assigned team, and submission and resolution timestamps.
4. Enable NLP/ML-based classification of tickets and prediction of urgency, with support for agent feedback to correct and improve predictions.
5. Provide real-time notifications and alerts to help support teams identify and respond quickly to urgent or critical tickets.
6. Maintain secure and efficient storage, retrieval, and management of ticket data, classification results, and historical support records.

7. Allow users to view, analyze, and export ticket data, classification results, and support reports in suitable formats such as CSV, JSON, and report formats.
8. Apply key software engineering principles such as security, scalability, maintainability, reliability, usability, and performance, while demonstrating proper requirements analysis, system design, SDLC selection, risk evaluation, limitations, and future enhancements.